

DO-IT-YOURSELF

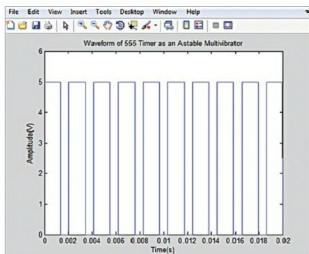


Fig. 3: Waveform for R1=1000 ohms, R2=1000 ohms and C=1μF

The duty cycle of the square waveform = $T_h / (T_h + T_f) \times 100\% = (T_h / T) \times 100\%$

The user must enter the values of R1, R2 and C, and click 'Run' in order to simulate the astable mode operation of 555 timer IC. On click-

ing 'Run' button, MATLAB executes a callback function to calculate and display the time period, frequency and duty cycle, and plots the waveform. The waveforms for two different sets of component values are shown in Figs 2 and 3, respectively.

You can also try out the following by building upon this work:

1. Plot voltage variation across capacitor C
2. Change the supply voltage from 5V to 10V and plot the output waveform; you will find that the amplitude of the output varies
3. Connect a diode (anode to pin 7 of IC and cathode to pin 6) in

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

parallel with R2 and plot the output waveform; you get a 50% duty-cycle output waveform **EFY**

Shibendu Mahata is M.Tech (gold medalist) in instrumentation and electronics engineering from Jadavpur University. Currently, he is pursuing Ph.D from NIT, Durgapur. He has keen interest in MCU-based real-time embedded signal processing and process control systems



Saikat Patra is passionate about electronics and MCU-based embedded system applications

Get a **FREE** IoT Starter Kit with 10 years subscription of EFY's ezine

MRP: ₹7000
Offer Price: ₹3500
You Save: ₹3500



Worth ₹3600



Worth ₹3400

Order Online: <http://newpro.kitsnspares.com>

Name _____ Designation _____ Organisation _____
Mailing Address _____ City _____
Pin Code _____ State _____ Phone/Mobile _____ Email _____ I would like to subscribe

Ezine Version of Electronics For You starting with the next issue. Please find enclosed a sum of Rs _____ by DD/MO/crossed cheque bearing the No. _____ dt. _____ in favour of EFY Enterprises Pvt Ltd, payable at Delhi. (*Please add Rs 50 on non-metro cheque)

Send this filled-in form or its photocopy to: EFY Enterprises Pvt Ltd D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110 020 | Ph: 011-26810601-03 | Fax: 011-26817563 | e-mail: info@efy.in | www.efy.in